

Interactive Digital Art and Design SE608

Proof of Concept - Concept Alignment Year IV

Student Name:	Liam Treacy	Student Number:	C00298242
Working Title:	Hank's Odyssey		
PoC Date:	01/10/2026	PoC Version:	Version 01
How to use this document:	<i>Replace the grey guidance text with project-specific content. This Proof of Concept is intended to show whether the proposed FYP idea is coherent and achievable. The document should demonstrate a clear link between the project theme, objectives, user needs, features, tools/technologies, implementation approach, expected output and evaluation. Keep the scope focused on proving the concept rather than developing the complete final project.</i>		
1. Project Theme / Problem:	The title of this project is Hank's Odyssey. The theme of the game is someone undertaking a grand journey and the resulting adventure. The protagonist, Henry "Hank" Quincy, leaves his home which has been left in ruin, to embark on a journey for a better life. The game creates an opportunity to tell a memorable story and feature some beautiful 3D environments.		
2. Project Aim:	To develop a fully completed 3D platforming game, which will feature an intriguing story and unforgettable characters with memorable voices.		
3. Project Objectives:	01: Create a strong visual identity with both 2D and 3D assets 02: Create prototypes and continually improve Hank's Odyssey using UPBGE and python, each iteration of the project will be an improved version. 03: Create a bug free game which runs perfectly 04: Have mechanics such as combat, player movement and dialogue interactions 05: Have a complete game, an executable which will allow users to play the game		
4. Target Users / Audience:	The target audience for Hank's Odyssey will be anyone 12 year olds and up. Choosing an older range will allow for themes like death and having features such as mild swearing and blood can be included. The game will be aimed for fans of 3D platformers and designed for owners of both computer and console.		
5. Main User Requirements / Use Cases:	01: Player starts game 02: Player clicks start button to begin playing 03: The opening cutscene will play, the player will have the option to skip 04: The player is given a choice to select 05: Gameplay begins 06: Player uses controller stick to move 07: Player hits jump button to jump		

6. Required Features / Functional Requirements:	. 2D Illustrated Cutscenes . Voice acting (In and out of cutscenes) , Movement and jumping in the 3D world . Boss Fight . 3D environments which can be explored . The choice at the beginning of the game to determine what path the player will take . Switching characters . 3 different stages . Executable build																												
7. Proposed Tools and Technologies:	<table><tr><th>Tool / Technology</th><th>Purpose in Project</th><th>Feature / Objective Supported</th><th>Why Suitable?</th></tr><tr><td>Blender</td><td>Creating 3D Models and Animations</td><td>3D Models needed to help build the world</td><td>Blender will be the back bone of this project, a lot of the game will be made within the Blender application, 3D models will be needed to create environments for the world</td></tr><tr><td>UPBGE</td><td>Be able to programme the game</td><td>Write Python Code to give the game functionality and meet the technical requirements</td><td>The game needs code, so this is how I will write code for my game, it is also very convenient because a majority of the game can be created in Blender, downloading and setting up UPBGE is a breeze.</td></tr><tr><td>Audacity</td><td>Record Audio</td><td>Having audio files which can be used in game for voice acting</td><td>Since I want to tell an engaging story in my game, I want to have voice acting to help tell the story, characters voiced by me and potentially other friends,</td></tr><tr><td>Sketchbook/Adobe Photoshop</td><td>2D Illustrations</td><td>Creating 2D cutscenes and textures</td><td>I have Sketchbook on my laptop, along with Photoshop, I wanted something to create 2D illustrations which will be how I do my cuisenes</td></tr><tr><td>Github/Git</td><td>Store progress of the project</td><td>Storing Code</td><td>I need somewhere to store my progress and leave a digital footprint of the project, Github and Git is perfect for this</td></tr></table>					Tool / Technology	Purpose in Project	Feature / Objective Supported	Why Suitable?	Blender	Creating 3D Models and Animations	3D Models needed to help build the world	Blender will be the back bone of this project, a lot of the game will be made within the Blender application, 3D models will be needed to create environments for the world	UPBGE	Be able to programme the game	Write Python Code to give the game functionality and meet the technical requirements	The game needs code, so this is how I will write code for my game, it is also very convenient because a majority of the game can be created in Blender, downloading and setting up UPBGE is a breeze.	Audacity	Record Audio	Having audio files which can be used in game for voice acting	Since I want to tell an engaging story in my game, I want to have voice acting to help tell the story, characters voiced by me and potentially other friends,	Sketchbook/Adobe Photoshop	2D Illustrations	Creating 2D cutscenes and textures	I have Sketchbook on my laptop, along with Photoshop, I wanted something to create 2D illustrations which will be how I do my cuisenes	Github/Git	Store progress of the project	Storing Code	I need somewhere to store my progress and leave a digital footprint of the project, Github and Git is perfect for this
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8. Project Alignment Matrix:	<i>This is the key section. Show the logical connection from each project objective to a required feature, the technology used to deliver it, how the PoC will provide evidence, and what the final project should produce.</i> <table><tr><th>Project Objective</th><th>Required Feature</th><th>Tool / Technology</th><th>PoC Evidence / Test</th><th>Expected Final Outcome</th></tr><tr><td>Create a strong visual identity with both 2D and 3D assets</td><td>2D illustrated cutscenes with character designs in both 2D and 3D.</td><td>Blender, Sktechbook</td><td>Create a sample character design in 2D and 3D, along with a short sample illustrated cutscene.</td><td>A consistent visual identity across the game's 2D illustrations, 3D characters and environments.</td></tr><tr><td>02: Create prototypes and continually improve Hank's Odyessy using UPBGE and python, each iteration of the</td><td>3D movement in 8 directions, player jumping and player interactions,</td><td>UPBGE, Python, Blender, Github/Git</td><td>Create an early playable prototype where the player can move, jump and interact with objects. Compare later versions against the</td><td>A fully playable game developed through multiple iterations, with improvements made to gameplay and functionality throughout</td></tr></table>					Project Objective	Required Feature	Tool / Technology	PoC Evidence / Test	Expected Final Outcome	Create a strong visual identity with both 2D and 3D assets	2D illustrated cutscenes with character designs in both 2D and 3D.	Blender, Sktechbook	Create a sample character design in 2D and 3D, along with a short sample illustrated cutscene.	A consistent visual identity across the game's 2D illustrations, 3D characters and environments.	02: Create prototypes and continually improve Hank's Odyessy using UPBGE and python, each iteration of the	3D movement in 8 directions, player jumping and player interactions,	UPBGE, Python, Blender, Github/Git	Create an early playable prototype where the player can move, jump and interact with objects. Compare later versions against the	A fully playable game developed through multiple iterations, with improvements made to gameplay and functionality throughout									
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9. PoC Demonstration / Evidence:	A small prototype will be created in UPBGE using Python, containing a playable character who can move in 8 directions, jump and interact with objects within a simple 3D environment. Basic gameplay elements such as dialogue and combat will also be added to show how the main gameplay systems will work together. This will also test the game's visual and storytelling elements by creating a small 2D illustrated cutscene with voice acting. This will demonstrate how the narrative will be integrated into the gameplay. The prototype will then be combined into a small playable demo to test the overall workflow and identify any technical issues before full development begins. The prototype will be a great starting point Hank's Odyssey. Final year project 01: Asset Creation Create initial 2D and 3D assets Design a protagonist and villain Create moodboards of how environments should look  02: Prototype Player movement in 8 directions Add character jumping Testing basic additions such as dialogue  03: Gameplay Systems Add basic necessities such as combat Test for bugs and refine gameplay  04: Story Telling Begin with 2D storyboards, later turned into cutscenes. Add voice acting  05: Demo Combine all previous elements to make a demo Present current progress to ensure project is on track  06: Complete Project Have a bug free game which has been thoroughly tested Create an executable 																				
10. Technology Suitability / Alternatives:	<table><tr><th>Requirement</th><th>Proposed Approach / Tool</th><th>Alternative Considered</th><th>Reason for Final Choice</th></tr><tr><td>3D Game Engine</td><td>UPBGE</td><td>Unity</td><td>UPBGE allows me to create the whole game within Blender, so it would save hopping from software to software. It also uses Python rather than C#.</td></tr></table>				Requirement	Proposed Approach / Tool	Alternative Considered	Reason for Final Choice	3D Game Engine	UPBGE	Unity	UPBGE allows me to create the whole game within Blender, so it would save hopping from software to software. It also uses Python rather than C#.									
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				which I am more skilled in.
	3D Modelling Software	Blender	Maya	Once again, I will be able to create all my project within Blender, I am also more comfortable using Blender than Maya.
	2D Drawing Software	Sketchbook	Krita/Adobe Photoshop	I need somewhere to draw my 2D cutscenes, I like Sketchbook and it's interface so I will try use that.
	Audio Recording and Editing Software	Audacity	Ocenaudio	Audacity is free and cross platform and is generally seen as the superior option, so I would prefer to use this
11. Scope - Core vs Optional Features:				
	Feature	Priority	Reason / Objective Supported	Include in PoC?
	3D Platforming Gameplay (Core)	Highest	The 3D gameplay is the backbone of the project. Without this, my project is not very substantial	Yes
	Give the player a choice of running away or hiding (optional)	Medium	The idea is that the game will begin with a choice, so the player can either stay and be killed by the assassin, or run and begin their journey To me this seems like a simple addition which will look more impressive than it is and help the game stand out.	Yes
	2D Illustrated Cutscenes with Voice Acting, including an opening and ending (Core)	High	The game needs some sort of visuals and cutscenes too sell the game. So I want to have 2D comic book style, and characters who	Yes
	Second playable character (core)	Low	The idea is an one point, after the boss battle, the final stage you switch between 2 characters and use their skills to complete the stage. The idea is a simple switch where you swap to another character and the model changes, with a little intermission like "Alright, you gotta take over here"	It has not been included yet, I want to try have it as an additional bonus and it is not a requirement.
	World map and ability to replay levels	Low	A small feature, the game is linear but after the player sees the ending, they can replay any level they choose.	Not included
12. Feasibility Assessment:				
	Area	Question to Consider		Assessment / Evidence
	Project Technical Aspect	Can the required features be built with the chosen technologies?		The core features such as 3D movement, jumping, combat, dialogue and character switching can be developed using UPBGE, Python and Blender. A PoC will be used to test the most technically important mechanics before full development.

	Time for Development	Can a meaningful version of the game be completed within the project timeframe?	The project will be developed in stages, beginning with a basic prototype before adding environments, gameplay systems, cutscenes and audio. Optional features can be removed if time becomes limited.																									
	Technical Complexity	Are any parts of the project likely to be particularly difficult to develop?	Features such as the boss fight, character switching and integrating voice acting with cutscenes may require more development time. These will be developed separately and tested as smaller prototypes before being added to the full game.																									
	Resources	Are the required software, hardware and other resources available?	Blender, UPBGE, Python, Sketchbook, Photoshop, Audacity and Git/GitHub are available for the project. A computer capable of running Blender and UPBGE will be required to develop and test the game.																									
	Scope	Can the project be kept to a realistic and manageable size?	The core project will focus on three playable stages, movement, combat, a boss fight, cutscenes and voice acting. Features such as level replay and a second playable character will be treated as optional and can be removed if necessary.																									
13. Risks and Plan B:																												
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14. PoC Outcome / Decision:	Select one outcome and justify the decision below: ☐ Proceed - the concept and proposed approach are suitable. ☐ Proceed with changes - the concept is feasible, but changes are required. ☐ Reconsider - the current concept or approach requires significant revision. Justification: [Explain the decision in 3-5 sentences.]								
15. Required Changes Before Full Development:	List the specific changes that must be made as a result of the PoC. Examples: change technology, reduce scope, replace an input method, redefine a feature, modify the user flow, add a Plan B, or strengthen the technical component.								
16. Next Steps:	List the next 3-5 concrete development tasks in priority order. Each task should follow logically from the PoC decision and the revised project direction.								
17. Supporting Material / References:	Provide links or references to supporting research, technical documentation, comparable systems, design references, GitHub repository, screenshots, diagrams or short demonstration videos used to justify the PoC decisions.								
Student declaration: The information in this Proof of Concept reflects the current proposed direction of the FYP and the evidence available at this stage. Signed: _____ Date: _____									
For Office / Supervisor Use Only									
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